

SkillNest

Agile Project Planning & Scrum Execution Report

Product Backlog, Sprint Planning, Scrum Ceremonies & Framework Evaluation for a Campus Placement & Internship Readiness Platform

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Course / Unit:	DevOps Engineering (Unit 2 Mini Project)	Submission Date:	September 06, 2026

1. Recap — Business Problem & Product Vision

Unit 1 established the Product Discovery foundation for SkillNest, the Campus Placement & Internship Readiness Platform. Before translating the platform into Agile deliverables, the Unit 1 artifacts were reviewed in full so that every Epic and User Story in this report stays grounded in a validated student and Training & Placement Officer (TPO) need.

- **Product Vision & Mission Statements:** reviewed to confirm the platform's purpose — giving every student a single place to build a verified profile, discover eligible drives, and track applications end-to-end.
- **Stakeholders:** Students, Training & Placement Officers (TPO), Recruiting Companies, and Institute Administration, as documented in the Unit 1 stakeholder matrix.
- **Business & Functional Requirements:** the BRD's four core workflows and six functional requirement items (FR-01 to FR-06) were mapped forward into the Epics below.
- **Non-Functional Requirements:** data privacy, availability, and performance targets (NFR-01 to NFR-03) were retained as Definition-of-Done constraints for every Sprint.
- **User Personas:** Aarav Joshi (final-year student), Mrs. Snehal Pawar (TPO officer), and Karan Shetty (recruiter) continue to anchor acceptance criteria in real user needs.
- **MVP & Roadmap:** the Unit 1 MVP scope and 4-phase engineering roadmap were used as the basis for the Sprint-level roadmap in Section 14.

2. Agile Product Vision

“Build a single, reliable platform that helps every student prepare for placements, discover the drives they are eligible for, and track each application — while giving the Training & Placement Office one accurate, real-time view of campus-wide placement activity.”

Vision Element	Definition
Product Objective	Replace scattered spreadsheets, notice-board circulars, and email threads with one role-aware digital platform for placement activity.
Primary Users	Students, TPO staff, recruiting company coordinators, and the platform's DevOps/technical team.
Business Benefits	Faster drive-to-application turnaround, fewer missed deadlines, audit-ready placement records, and higher-quality student-company matching.
Success Criteria	99.5% uptime, drive alerts delivered within 5 minutes of posting, application status updates reflected within 1 hour, zero duplicate applications.
Project Constraints	4-Sprint MVP timeline, 5-member student development team, cloud-hosted PostgreSQL backend, curriculum-fixed submission dates.

Vision Element	Definition
Expected Capabilities	RBAC authentication, profile & resume management, drive posting and eligibility filtering, application tracking, mock-interview practice, notifications, and reporting.

3. Scrum Team Structure & Roles

For academic delivery, a compact 5-member Scrum Team was defined. Akash Suresh Dhanwate serves as Product Owner, with peer roles simulated to complete the team structure required for realistic Scrum execution.

Team Member	Scrum Role	Key Responsibilities
Akash Suresh Dhanwate	Product Owner	Defines requirements, prioritizes the Product Backlog, represents stakeholder interests, accepts completed work.
Riya Kulkarni	Scrum Master	Facilitates ceremonies, removes impediments, ensures Agile practices are followed, supports collaboration.
Omkar Jadhav	Developer — Backend	Designs APIs and the database schema, implements core services, writes unit tests.
Sneha Wagh	Developer — Frontend	Builds student and TPO portals, integrates REST APIs, prepares component-level documentation.
Tanvi Kale	QA / Developer	Writes and executes test cases, validates acceptance criteria, demonstrates completed features in Sprint Review.

3.1 Role Responsibility Summary

- **Product Owner:** owns the “what” — requirement definition, backlog prioritization, stakeholder representation, and story acceptance.
- **Scrum Master:** owns the “how well” — ceremony facilitation, impediment removal, and coaching the team in Agile practice.
- **Development Team:** owns the “how” — design, implementation, testing, documentation, and Sprint Review demonstration.

4. Epic Documentation

The complete SkillNest application was decomposed into seven high-level Epics, each mapped to the Unit 1 functional requirements.

Epic	Purpose & Scope	Key Features
Epic 1 — Student Profile & Resume	Own the full lifecycle of a student's profile and resume data.	Profile Creation, Skill Tagging, Resume Upload, Resume Builder
Epic 2 — Company & Drive Management	Maintain the company catalogue and job/internship drive postings.	Add Company, Post Drive, Eligibility Criteria, Drive Calendar
Epic 3 — Application Tracking	Track each student's applications across every drive.	Apply to Drive, Track Status, Withdraw Application, Application History
Epic 4 — Mock Interview & Assessment Practice	Provide practice tools ahead of real interviews.	Aptitude Test Bank, Mock Interview Scheduling, Feedback Capture, Readiness Score
Epic 5 — Authentication & Security	Secure every role's access to the platform.	User Login, Logout, Password Reset, Role-Based Access Control
Epic 6 — Notifications	Keep students and TPO staff proactively informed.	Drive Alerts, Deadline Reminders, Result Notifications
Epic 7 — Reporting & Analytics	Give the TPO consolidated, exportable placement insights.	Placement Reports, Drive-wise Reports, Student Readiness Reports, Company-wise Reports

5. User Story Repository

Every Epic was decomposed into User Stories using the standard “As a ... I want ... so that ...” format. A representative set from the repository is shown below; the full repository contains 40+ stories across all seven Epics.

ID	Epic	User Story
US-01	Authentication	As a registered user, I want to log in securely so that I can access role-appropriate features.
US-02	Authentication	As a user, I want to reset my password so that I can regain access if I forget my credentials.
US-03	Student Profile	As a Student, I want to create my profile with academic details so that recruiters see accurate information.
US-04	Student Profile	As a Student, I want to upload and update my resume so that it is always current for applications.
US-05	Company & Drive Management	As a TPO Admin, I want to add a company and post a drive so that eligible students can apply.
US-06	Company & Drive Management	As a TPO Admin, I want to define eligibility criteria for a drive so that only eligible students can apply.
US-07	Application Tracking	As a Student, I want to apply to a posted drive so that my interest is recorded.
US-08	Application Tracking	As a Student, I want to track my application status so that I know where I stand.
US-09	Mock Interview Practice	As a Student, I want to take a mock aptitude test so that I can gauge my readiness.
US-10	Mock Interview Practice	As a Student, I want to schedule a mock interview so that I can practice before the real one.
US-11	Notifications	As a Student, I want to receive an alert when a new drive matching my profile is posted so that I don't miss it.
US-12	Reporting	As a TPO Admin, I want to generate a placement report so that I can review overall outcomes.

6. Acceptance Criteria Document

Acceptance Criteria define measurable conditions of “Done” for each User Story. Two representative examples are detailed below.

6.1 User Story: Student Login (US-01)

- Login page loads successfully within 1.5 seconds on standard broadband.
- Registered users can log in using their institutional email and password.
- Invalid credentials display a clear, actionable error message.
- A successful login redirects the user to their role-specific dashboard.
- Session state is managed securely and expires after a defined idle period.

6.2 User Story: Apply to Drive (US-07)

- Students can view only the drives for which they meet the eligibility criteria.
- A student can apply to a drive only once; duplicate applications are blocked.

- The application is timestamped and immediately visible to the TPO Admin.
- The student receives an on-screen and notification confirmation after applying.
- The application appears in the student's Application History with status "Submitted."

7. Prioritized Product Backlog

User Stories were consolidated into a single Product Backlog and ranked by business value, user importance, technical dependency, and implementation complexity.

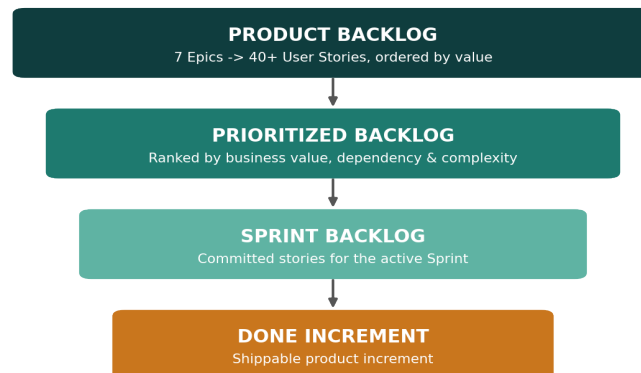


Figure 7.1 — Backlog refinement funnel from Epics to a shippable increment

Priority	Epic	User Story	Story ID
High	Authentication	User Login	US-01
High	Student Profile	Create Profile	US-03
High	Company & Drive Management	Post Drive	US-05
High	Application Tracking	Apply to Drive	US-07
Medium	Company & Drive Management	Eligibility Criteria	US-06
Medium	Reporting	Placement Report	US-12
Medium	Notifications	Drive Alert	US-11
Low	Mock Interview Practice	Schedule Mock Interview	US-10

8. Story Point Estimation Sheet

The team estimated effort using Planning Poker, with a modified Fibonacci scale.

Story Points	Complexity
1	Very Easy
2	Easy
3	Moderate
5	Difficult
8	Very Difficult
13	Highly Complex

Story ID	User Story	Story Points
US-01	User Login	3
US-02	Password Reset	2
US-03	Create Profile	3
US-04	Upload Resume	2
US-05	Post Drive	3
US-06	Eligibility Criteria	5
US-07	Apply to Drive	3
US-08	Track Application Status	2
US-09	Aptitude Test	5
US-10	Schedule Mock Interview	5
US-11	Drive Alert	3
US-12	Placement Report	5

9. Sprint Planning

The Product Backlog was organized into four two-week Sprints aligned with the Unit 1 engineering roadmap.

Sprint	Sprint Goal	Key Activities
Sprint 1	Develop user authentication and complete project setup.	Repository initialization, login, password reset, base UI shell, initial test suite
Sprint 2	Develop the Student Profile & Resume module end-to-end.	Create Profile, Upload Resume, Update Profile, Skill Tagging
Sprint 3	Develop Drive Management and Application Tracking.	Add Company, Post Drive, Eligibility Criteria, Apply to Drive, Track Status
Sprint 4	Develop Mock Interview Practice, Notifications, Reporting, and prepare the MVP release.	Aptitude Test, Schedule Mock Interview, Drive Alerts, Placement Report, release hardening

9.1 Sprint 1 Backlog (Illustrative)

Story ID	User Story	Points	Owner
US-01	User Login	3	Omkar Jadhav
US-02	Password Reset	2	Omkar Jadhav
—	Repository & CI scaffolding	5	Sneha Wagh
—	Base UI shell & navigation	5	Sneha Wagh
—	Sprint 1 test plan & execution	3	Tanvi Kale

10. Scrum Board

The team maintained a five-column Scrum Board to track story status throughout each Sprint. A snapshot from mid-Sprint 1 is shown below.

Backlog	To Do	In Progress	Testing	Completed
Post Drive	Create Profile	Login UI	CI Pipeline	Repository Setup
Apply to Drive	Upload Resume	Password Reset Flow	—	Base UI Shell
Eligibility Criteria	—	—	—	—

11. Burndown Chart

Sprint 1 committed 18 story points across a 10-working-day Sprint. Actual progress tracked slightly behind the ideal line mid-Sprint due to environment setup delays, then converged by Sprint end.

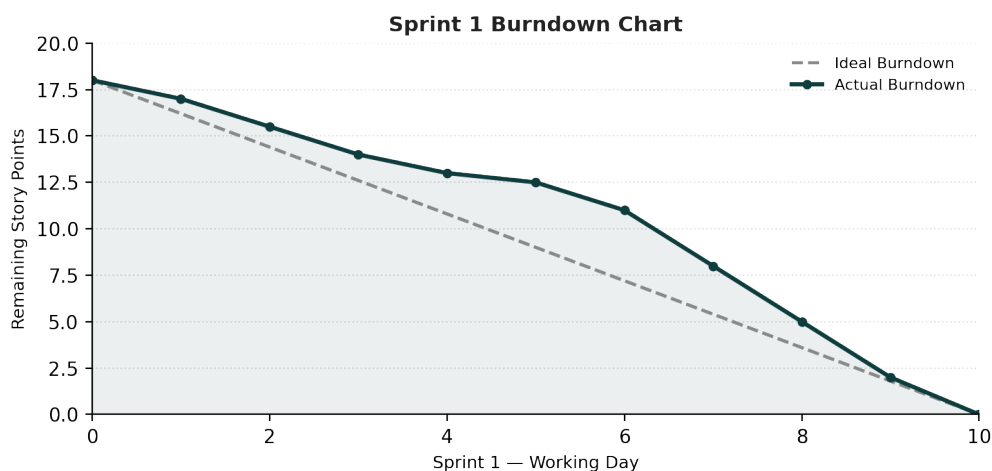


Figure 11.1 — Sprint 1 ideal vs. actual burndown

12. Simulated Scrum Ceremonies

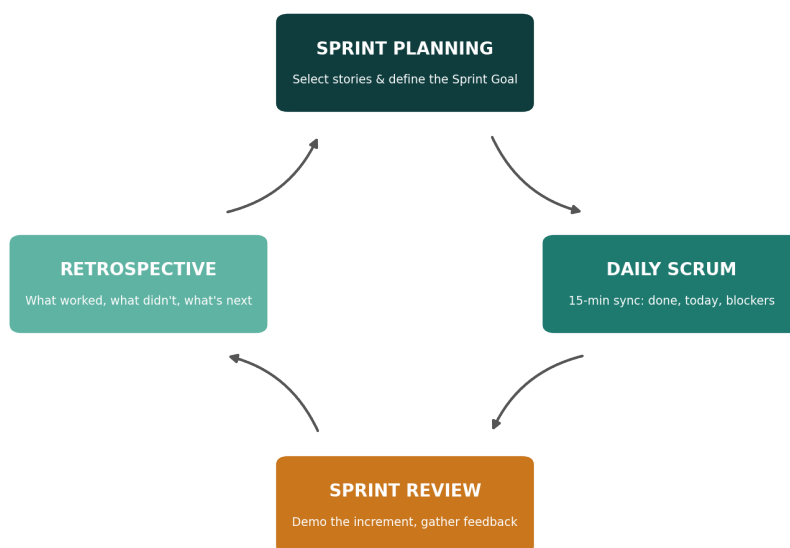


Figure 12.1 — The simulated Sprint cycle of Scrum ceremonies

12.1 Sprint Planning

The team selected US-01, US-02, and supporting setup tasks for Sprint 1, estimated effort via Planning Poker, and agreed the Sprint Goal: "Develop user authentication and complete project setup."

12.2 Daily Scrum

Each of the 10 working days included a 15-minute stand-up where every member answered: what was completed yesterday, what will be done today, and what impediments exist. Riya Kulkarni (Scrum Master) tracked and resolved blockers, such as a delayed database provisioning task on Day 3.

12.3 Sprint Review

At the end of Sprint 1, the Development Team demonstrated working login and password-reset flows to the Product Owner. Feedback requested clearer inline validation messages, which was added to the Sprint 2 backlog.

12.4 Sprint Retrospective

What Went Well	Challenges Faced	Improvements for Next Sprint
Repository and CI pipeline were ready on Day 1	Database provisioning took longer than estimated	Pre-provision infrastructure before Sprint Planning
Daily Scrums stayed within 15 minutes	Story point estimates for auth flows were slightly optimistic	Add a buffer for security-related stories
Clear Definition of Done reduced rework	Limited cross-testing between frontend and backend	Introduce a shared integration-testing session mid-Sprint

13. Agile Framework Comparison

Framework	Best Suited For	Fit for SkillNest
Scrum	Small, cross-functional teams with fixed-length iterations	Strong fit — matches the 4-Sprint academic timeline and small team size
Kanban	Continuous flow work with variable priorities	Useful for production support after MVP release, not initial build
Extreme Programming (XP)	Teams needing strong engineering discipline (TDD, pairing)	Complements Scrum for the eligibility-matching and authentication modules
Lean Development	Eliminating waste and maximizing flow efficiency	Principles applied to backlog refinement, not a standalone framework here
Large Scale Scrum (LeSS)	Multiple Scrum teams on one product	Not required — team size is a single 5-member unit
Scaled Agile Framework (SAFe)	Enterprise programs with many coordinated teams	Over-engineered for a single-team academic project
Scrum@Scale	Scaling Scrum across many teams via a Scrum of Scrums	Not applicable at current team size

Recommendation: Scrum is the most suitable framework for SkillNest given the project's small team size, fixed academic timeline, and need for frequent stakeholder feedback. XP practices (automated testing, continuous integration) are adopted within Scrum Sprints to strengthen code quality on security- and eligibility-matching stories.

14. Project Timeline & Release Roadmap

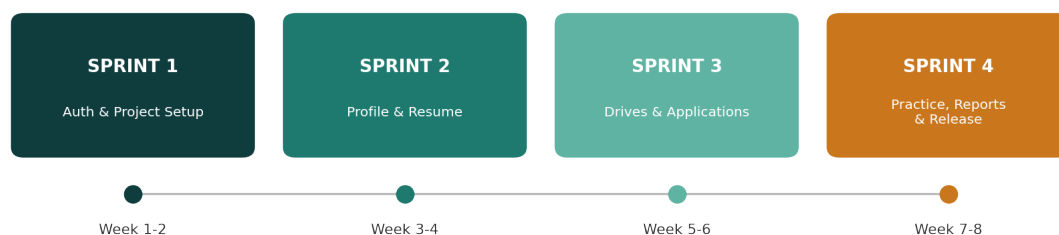


Figure 14.1 — 4-Sprint release roadmap toward the MVP

15. Deliverables Verification & Sign-Off

All Unit 2 deliverables specified in the curriculum have been completed in full:

- Agile Project Planning Report
- Product Vision Document
- Scrum Team Structure
- Scrum Roles and Responsibilities Document
- Epic Documentation (7 Epics)
- User Story Repository
- Acceptance Criteria Document
- Prioritized Product Backlog
- Story Point Estimation Sheet
- Sprint Planning Document (4 Sprints)
- Sprint Backlog (Sprint 1 illustrative)
- Scrum Board
- Burndown Chart
- Sprint Review Report
- Sprint Retrospective Report
- Agile Framework Comparison Report
- Project Timeline and Release Roadmap

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